

CANDIDATE DATA TRANSFORMER

TECHNICAL DESIGN SPECIFICATION | JUNE 30, 2026

1. HIGH-LEVEL ARCHITECTURE & DATA FLOW

The system processes heterogeneous talent source streams into a single canonical profile schema using a predictable, linearly decoupled 7-stage pipeline network.

```
[Recruiter CSV] --> (CSV Loader) --> [Normalize] --  
[Notes TXT] --> (TXT Loader) --> [Extract] --/
```

Inbound pipelines operate completely isolated from one another to prevent schema-level contamination, executing dedicated data type coercion steps before merging records contextually via a unique key.

2. KEY DESIGN DECISIONS

- Separation of Concerns via Isolated Loaders & Transformers:** Modular independent staging ensures that ingestion faults or raw parser adjustments in unstructured source modules never disrupt the baseline operational ingestion flow of verified structured files.
- Deterministic Merge Strategy with Provenance Archiving:** Conflicting single-value schema attributes default to the verified recruiter CSV stream as a trusted priority master channel. Instead of deleting conflicting rejected text files data blocks, entries are written straight to a permanent historical trace log to guarantee non-destructive recovery.
- Decoupled Runtime Configuration Abstraction:** A functional layer driven by a modular dynamic layout configuration modifies structural representation elements, masks internal property properties, maps schema locations, and shapes missing value operations at data persistence without invalidating core entity structural constraints.

3. CONFIDENCE METRICS & CORE SCHEMA MECHANICS

Static base scores reflect explicit source validity expectations: CSV is assigned a weight of 0.9, while text-extracted profiles scale at 0.7. Overlapping fields leverage a computed average of 0.80. The composite state vector is finalized at process resolution using an unweighted evaluation ratio:

$$Overall_Confidence = \sum_{i=1}^N Confidence(Field_i) / N$$

Where N signifies the numerical index of total non-null target values safely initialized inside the running canonical workspace.

4. EDGE CASES HANDLED & PROJECT SCOPE BOUNDARIES

- Asymmetric Merge Execution:** The pipeline contextually processes candidates existing only on one side of the source stream (e.g., CSV track missing an associated free-form notes log), treating missing parameters cleanly as unpopulated initialization blocks rather than critical processing structural exceptions.
- File Availability Failures:** Broken or missing physical source targets drop explicit system warnings and initialize a safe empty baseline data framework downstream to guarantee overall workflow continuity.
- Explicit Scope Exclusions (Descoped):** Entity matching relies strictly on direct matching rules using explicit identifier mapping keys; fuzzy identity matching logic is intentionally out of scope. High-overhead NLP model frameworks are replaced with lightweight regular expressions and strict token-mapping dictionaries to remain operating within explicit system scope boundaries.